

Data Model — Ambula Healthcare Platform

9 MongoDB collections · Every field with data type & constraint · FK references · Cardinality

PK Primary Key

FK Foreign Key (ref)

● Regular field

Bold = field name Grey = data type

Dashed line + label = relationship

User	
_id	ObjectId
name	String · required
email	String · unique · required
password	String · required (bcrypt)
role	Enum: patient doctor admin
phone	String
language	String · default: English
createdAt	Date · auto
updatedAt	Date · auto

Doctor	
_id	ObjectId
userid	→ User · unique · required
specialization	Enum: General Cardiology Dermatology Pediatrics ...
qualifications	Array [(degree, institution, year)]
licenseNumber	String · unique · required
consultationFee	Number · min:0 · required
bio	String
experience	Number (years)
location	String
languages	Array [Enum: English Hindi Kannada Tamil ...]
workingHours	{startTime, endTime, daysOfWeek[0-6]}
averageRating	Number · 0-5 · default: 0
totalReviews	Number · default: 0
isAvailable	Boolean · default: true
profileImage	{field: ObjectId, url: String}
createdAt	Date · auto
updatedAt	Date · auto

Slot	
_id	ObjectId
doctorId	→ Doctor · required
appointmentId	→ Appointment (when booked)
date	Date · required
startTime	String · HH:MM · required
endTime	String · HH:MM · required
status	Enum: available booked locked blocked
consultationType	Enum: online offline both
maxPatients	Number · default: 1
bookedCount	Number · default: 0
lockTimestamp	Date (concurrency guard)
createdAt	Date · auto
updatedAt	Date · auto

Appointment	
_id	ObjectId
patientId	→ User · required
doctorId	→ Doctor · required
slotId	→ Slot · required
bookingId	String · unique · required
patientDetails	{name, age, phone} booking snapshot
status	Enum: scheduled completed cancelled no-show rescheduled
consultationType	Enum: online offline · required
consultationLink	String (online only)
notes	String
consultation	{diagnosis, notes, prescriptionId→, followUpDate}
payment	{amount, status, transactionId, paidAt}
statusHistory	Array [{status, timestamp, changedBy, reason}]
reminderSent	Boolean · default: false
createdAt	Date · auto
updatedAt	Date · auto

HealthSummary	
_id	ObjectId
userid	→ User · unique · required
bloodGroup	Enum: O+ O- A+ A- B+ B- AB+ AB-
height	Number (cm)
weight	Number (kg)
allergies	Array [{name, severity, reactions}]
medicalConditions	Array [{name, diagnosedDate, status}]
currentMedications	Array [{name, dosage, frequency, startDate, endDate}]
familyHistory	Array [{relation, condition, notes}]
emergencyContacts	Array [{name, phone, relation}] · required
vaccinations	Array [{name, date, nextDueDate}]
lastCheckupDate	Date
createdAt	Date · auto
updatedAt	Date · auto

Prescription	
_id	ObjectId
appointmentId	→ Appointment · required
doctorId	→ Doctor · required
patientId	→ User · required
prescriptionNumber	String · unique · required
diagnosis	String · required
medicines	Array [{name, dosage, frequency, duration, instructions}]
tests	Array [{name, purpose}]
followUpDate	Date
attachments	Array [{fileId, url, type, uploadedAt}]
qrCode	{data: String (dataURL), generatedAt: Date}
status	Enum: draft issued filled completed
createdAt	Date · auto
updatedAt	Date · auto

Waitlist	
_id	ObjectId
slotId	→ Slot · required
doctorId	→ Doctor · required
patientId	→ User · required
bookingIdIfAccepted	→ Appointment (on accept)
position	Number · required
status	Enum: waiting notified booked expired cancelled
addedAt	Date · auto
expiresAt	Date · TTL 7 days auto-delete
notificationSentAt	Date

HealthRecord	
_id	ObjectId
userid	→ User · required
relatedAppointmentId	→ Appointment
type	Enum: prescription lab-report scan medical-history vaccine-record
title	String · required
description	String
file	{fileId, url, fileName, mimeType, size}
tags	Array [String]
visibility	Enum: private shared-doctors
sharedWith	Array [{doctorId, sharedAt}]
createdAt	Date · auto
updatedAt	Date · auto

Notification	
_id	ObjectId
userid	→ User · required
type	

Schema Indexes & Constraints

Collection	Index	Type	Purpose
Doctor	{specialization:1, isAvailable:1}	Compound	Filter available doctors by specialty
Doctor	{averageRating:-1}	Single	Sort by rating (descending)
Doctor	{location:1}	Single	Location-based search
Slot	{doctorId:1, date:1, status:1}	Compound ★	Core booking query — CRITICAL
Slot	{date:1, status:1}	Compound	All available slots on a date
Appointment	{patientId:1, createdAt:-1}	Compound	Patient appointment history
Appointment	{doctorId:1, createdAt:-1}	Compound	Doctor appointment list
Appointment	{status:1, createdAt:-1}	Compound	Filter by status
Appointment	{slotId:1}	Single	Lookup appointment by slot
Prescription	{patientId:1, createdAt:-1}	Compound	Patient prescription history
Prescription	{doctorId:1, createdAt:-1}	Compound	Doctor-issued prescriptions
HealthRecord	{userId:1, createdAt:-1}	Compound	Records sorted by date
HealthRecord	{userId:1, type:1}	Compound	Filter by document type
Notification	{userId:1, createdAt:-1}	Compound	Inbox sorted by date
Notification	{userId:1, status:1}	Compound	Unread count query
Notification	{expiresAt:1}	TTL	Auto-delete after 30 days
Waitlist	{expiresAt:1}	TTL	Auto-delete after 7 days

Collection Summary

Collection	Fields	Unique Constraints	Cardinality
User	9	email	1:1 Doctor, 1:1 HealthSummary, 1:N rest
Doctor	17	userId, licenseNumber	1:N Slot, Appointment, Prescription
Slot	13	Compound (doctorId + date + status)	1:1 Appointment, 1:N Waitlist
Appointment	16	bookingId	1:1 Prescription, 1:N HealthRecord
Prescription	14	prescriptionNumber	1 per Appointment
Waitlist	10	TTL 7 days	N per Slot
HealthRecord	12	—	N per User, optional link to Appointment
HealthSummary	14	userId (one per patient)	1:1 User
Notification	11	TTL 30 days	N per User